

Unit 5 :: Fundamentals of Data Visualization

☞ SEC-03-PHY-91-R-03 ☜

Dr. Parag Bhattacharya
Department of Physics
Rangapara College (Autonomous)

Syllabus

Unit 5: Fundamentals of data visualization

Principles and use cases for categorical and continuous data plots, including column, line, pie, bar, area and scatter plots; customizing chart elements (axes, scales, legends, and labels); fine-tuning aesthetics for clarity and technical accuracy; advanced plotting techniques: 3D surface plots, contour mapping and multi-layered/multipanel figures.

1 Categorical and continuous data plots

A dataset is said to be *categorical* if its independent variable takes one of a small, discrete set of labels (e.g. the day of the week, the floor number of a skyscraper), and *continuous* if its independent variable can take any value within a range (e.g. time, angle, or magnetic field strength). The choice of plot type should follow from which of these two kinds of variable is being plotted, not from visual preference alone.

1.1 Column and bar plots

In a *column plot*, each category is represented as a vertical bar whose height is proportional to a summary value (such as a mean, a total or a count). In a *bar plot*, the idea is the same except that it is drawn using horizontal bars.

Both types of plots are appropriate when the independent variable is categorical. For example, they can be used to compare the mean temperature across different weeks.

1.2 Line plots

A *line plot* connects successive data points with straight (or smoothed) line segments. It is the standard choice for a continuous independent variable, most commonly time. It should not be used for categorical data, since a line drawn between two unrelated categories implies a continuity between them that does not exist.

1.3 Pie charts

A *pie chart* represents each category's share of a whole as a wedge of a circle, with wedge angle proportional to that category's proportion of the total. Pie charts are appropriate only for categorical data that sums to a meaningful whole, and are best limited to a small number of categories (three to five); beyond this, a column chart is usually clearer.

1.4 Area plots

An *area plot* is a line plot with the region between the line and the axis filled in. It is used, in place of an ordinary line plot, when the magnitude of a continuously-varying quantity is the point of emphasis, or when several such quantities are to be compared by their relative contribution to a stacked total.

1.5 Scatter plots

A *scatter plot* represents each observation as an individual point, with no line joining successive points. It is the correct choice when the relationship between two continuous variables is of in-

terest, rather than the trend of one variable against time. A scatter plot is particularly used when there is no reason to assume the points are ordered or continuous in the underlying physical process.